

Electromagnetic Compatibility: GROUNDING



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In the previous parts we covered multi-layer PCB designing. Now, let us look at grounding.

Grounding can be defined as a connection, whether intentional or accidental, between an electrical circuit or equipment and earth or to some conducting body that serves in place of earth. Hence, one can conclude that grounding is nothing but connection to earth or to a conductor that serves as earth. Going by this definition, a connection between a circuit and its 0V reference can also be considered as grounding, while earthing can be considered as a case of grounding where the ground is essentially planet Earth.

Purpose of grounding

The primary aim of grounding is to prevent shock hazard that exists in a high-voltage distribution system. Other functions include power fault clear-out, protection against lightning and electrostatic discharge (ESD) hazards and protection against electromagnetic interference (EMI).

Preventing shock hazard. Shock hazard exists when equipment enclosure/chassis, by virtue of a fault or otherwise,

is not connected to ground. If a live wire is accidentally connected to the chassis, then the latter becomes live. Now, if you happen to touch it, current will get a path to ground through your body. If the chassis is connected to ground, say, by a low-resistance wire, fault current would prefer to flow through this low-resistance path rather than your body. This current triggers a fuse or circuit breaker, alerting you. Even if you ignore this warning, the frame will be at ground potential and there will be no danger of a shock.

Power fault clear-out. Aging, insulation damage, environmental contaminants in insulation and so on can cause breakdown in insulation during a fault, resulting in arcing that can lead to fire. When the neutral terminal of power supply is connected to earth, any fault in grounding would show up as a phase-to-neutral short and would trip the fuse, arresting the hazardous voltage.

Protection against lightning hazard. Grounding is essential for draining severe lightning currents (30kA to 100kA) to ground, which otherwise can be life-threatening and can also damage electronic components in an equipment.

Fig. 18: Grounding hierarchy for a typical installation

